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A review article on the assessment of additive manufacturing

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Abstract

Additive manufacturing (AM), commonly known as 3D printing, has revolutionized the manufacturing landscape by enabling layer-by-layer fabrication of complex geometries from digital models. This paper provides a comprehensive overview of the evolution, current capabilities, and future directions of AM. Beginning with the historical rise of AM, it explores and compares its major technological categories, including material extrusion, vat photopolymerization, powder bed fusion, and directed energy deposition. Each technology is discussed with regard to standard classifications and operational mechanisms. It further examines the crucial role of material properties and selection, emphasizing how polymers, metals, ceramics, and composites influence mechanical performance and application suitability. The paper investigates the deployment of AM across industries such as aerospace, biomedical, automotive, construction, and consumer goods, highlighting transformative applications. Despite its benefits, AM faces challenges such as anisotropic mechanical properties, limited material diversity, high energy consumption, and scalability constraints. Recent advancements leveraging machine learning (ML) or (AI) integration are discussed, particularly in process monitoring, defect prediction, and print quality optimization. ML-integrated process optimization techniques are shown to enhance part performance and production efficiency. Additionally, this study compares AM with subtractive manufacturing (SM), focusing on material utilization, energy efficiency, and production flexibility. A life cycle assessment (LCA) is conducted to evaluate the environmental and economic impacts of AM technologies. Market analysis indicates substantial global growth of the AM industry, fueled by technological maturation and increasing demand for customized solutions. Finally, it projects future research directions, including the development of multi-material printing, integration of AI-driven adaptive systems, sustainable material innovations, and the role of AM in decentralized manufacturing. This holistic analysis affirms AM's pivotal role in reshaping the future of manufacturing with enhanced sustainability, precision, and design freedom. Overall, this review offers a big-picture view of AM where it stands today and how it's paving the way for a more innovative, sustainable, and flexible future in manufacturing.

Highlights

- Versatile Applications across Industries: Additive manufacturing enables complex geometries and rapid prototyping.
- Wide Range of AM Technologies: AM technologies vary in material compatibility and process efficiency.
- Material Diversity and Selection: Key challenges include surface finish, mechanical properties, and cost.
- Machine Learning for Smarter Manufacturing: AI-driven optimization enhances process control and material selection.
- Sustainable and Growing Future Outlook: Sustainability efforts focus on recycling and bio-based materials.

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• Comparative Overview of Additive vs. Subtractive Manufacturing.

Keywords Additive Manufacturing, 3D Printing, Process Optimization, AI Integration, Sustainable Manufacturing, Life Cycle Assessment

Introduction

Additive manufacturing (AM), commonly known as 3D printing, is an advanced fabrication technique that constructs components layer by layer from digital models. Over the past decades, AM has gained prominence due to its ability to produce complex structures with minimal material waste. The technology has applications across industries such as aerospace, healthcare, and automotive manufacturing. However, various challenges must be addressed to enhance efficiency and sustainability (Dehghan et al. 2025; Haleem and Javaid 2019a). This critical review delves into the rapidly expanding field of additive manufacturing (AM), also known as 3D printing. We will examine recent advancements, explore their diverse applications across various sectors, and critically analyze the associated challenges. The remarkable growth of AM is inextricably linked to the transformative demands of Industry. Several studies highlight this connection, emphasizing AM's role as a critical pillar in this industrial revolution (Haleem and Javaid 2019b; Khorasani et al. 2021). The integration of AM into the industry necessitates a thorough understanding of its capabilities and limitations, prompting a comprehensive analysis of its potential and its current shortcomings. This review will explore a range of AM technologies, assess their material compatibility and limitations, and investigate their impact on supply chain dynamics (Debnath et al. 2022). Furthermore, we will delve into the significant role of machine learning (ML) in optimizing AM processes and enhancing the overall quality of the manufactured products (Wu 2023; Ng et al. 2024a). The aim is to provide a nuanced perspective on the current state of AM, identifying both its remarkable achievements and the persistent hurdles that need to be overcome for its widespread and sustainable adoption (Figs. 1, 2 and 3).

The rise of Additive Manufacturing (AM)

Additive manufacturing has emerged as a revolutionary technology that has significantly impacted various industries, including aerospace, automotive, healthcare,

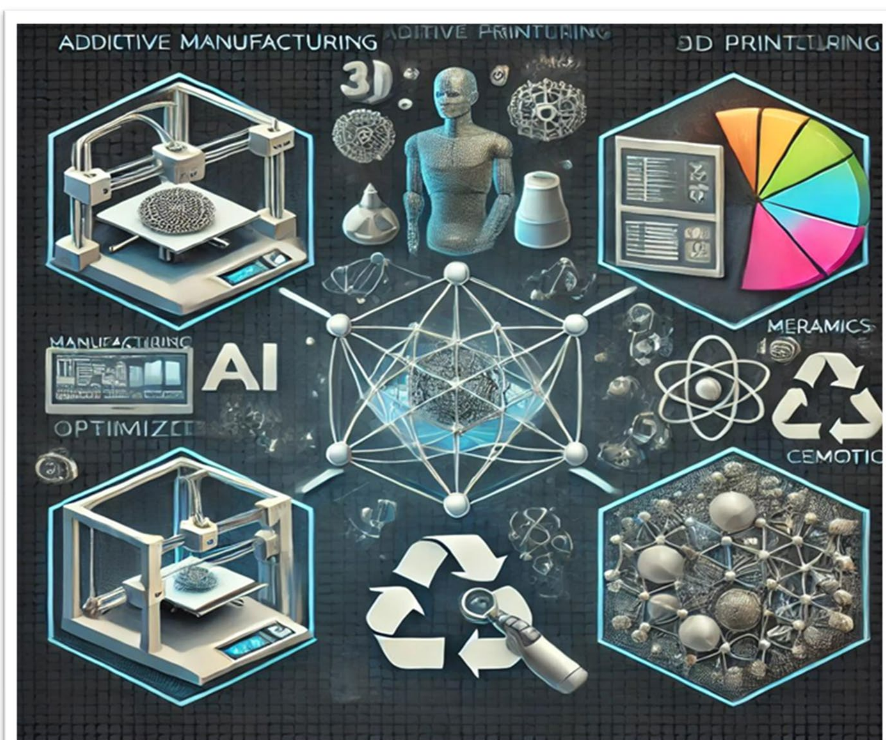

Fig. 1 The Graphical Abstract of Additive Manufacturing (AM)

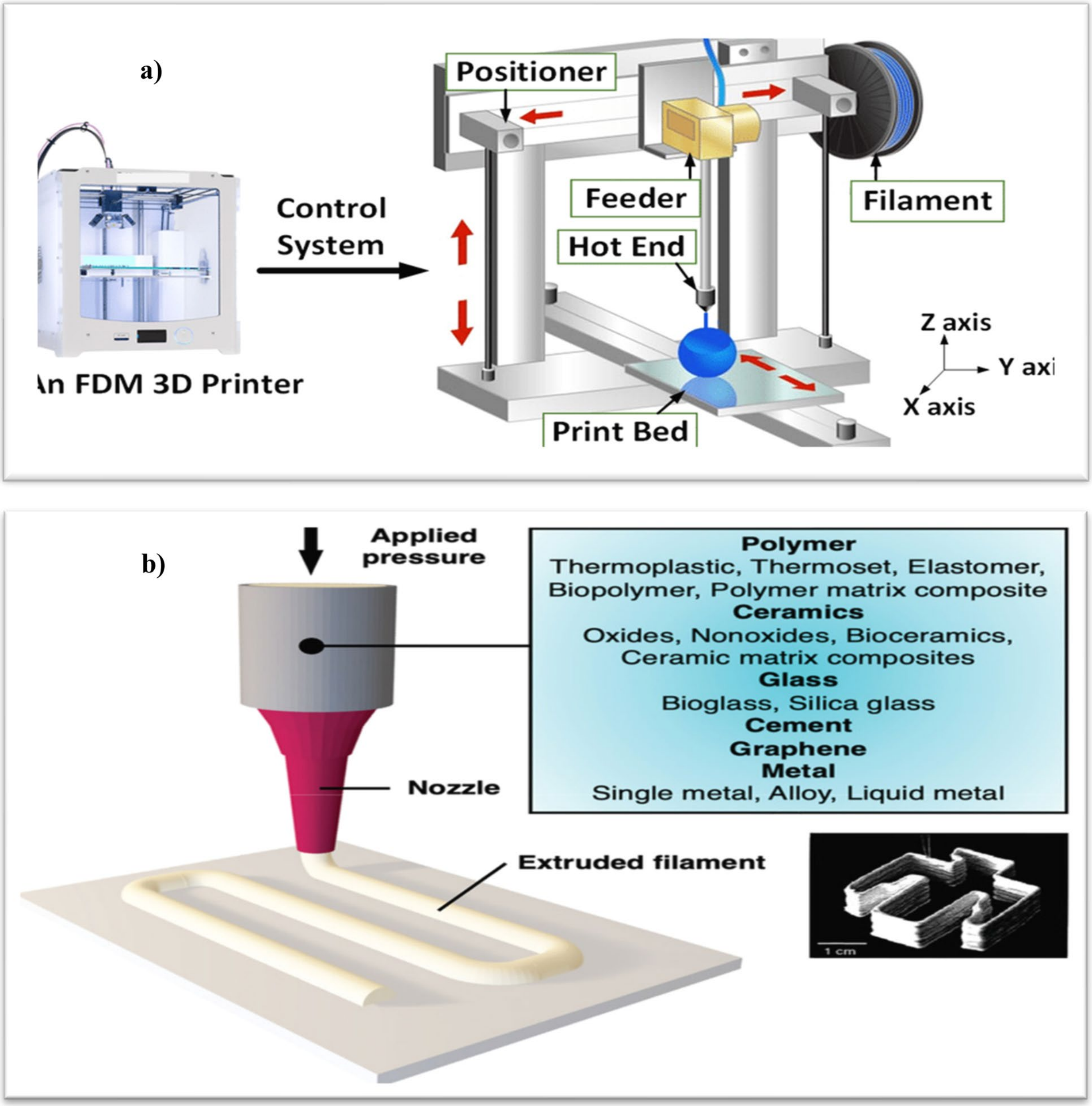

Fig. 2 a) Mechanism of a 3D printer with inevitable variations arising from three primary components: feeder, positioner, and hot end **b)** Extruded filament by applying pressure through the nozzle (Li et al. 2018)

and construction. AM refers to a family of manufacturing processes that fabricate objects layer by layer based on digital 3D models. Unlike traditional subtractive manufacturing methods, which involve cutting away material, AM builds up the material to create complex geometries that are difficult or impossible to achieve with conventional techniques (3D Printing/ Additive Manufacturing Technologies in Dentistry : A Bibliometric Study Diş Hekimliğinde 3D Baskı/ Katmanlı Üretim Teknolojileri 2025). This technology allows for the creation of highly customized parts, rapid prototyping, and complex structures with less material waste (Roscoe 2023). In recent years, AM has seen significant advancements in both materials and processes. The development of new materials, such as metal alloys, biocompatible materials, and composites, has expanded the scope of AM applications (Mondal, et al. 2024). This is coupled with the evolution of AM

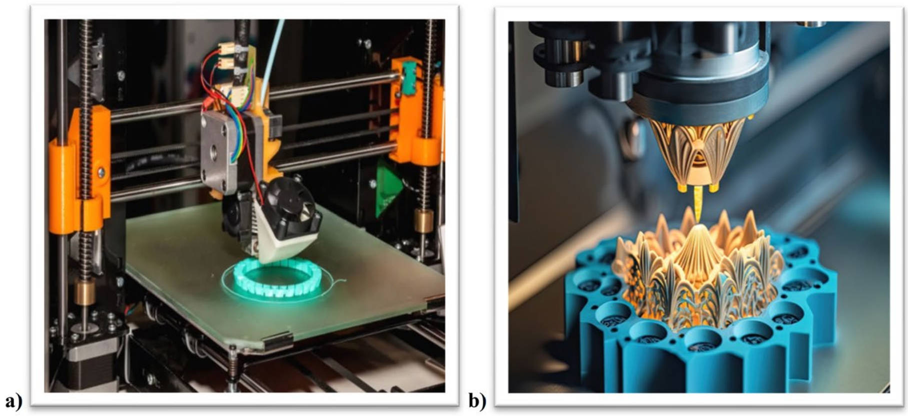

Fig. 3 **a)** Image of a practical additive manufacturing and **b)** Subtractive manufacturing

technologies such as selective laser sintering (SLS), stereolithography (SLA), and fused deposition modeling (FDM), which have become more accurate, efficient, and cost-effective. As a result, AM has begun to replace traditional manufacturing methods in high-precision sectors, where customization and rapid production are crucial (Jayswal and Adanur 2022). The aerospace and automotive industries have particularly embraced AM due to its ability to create lightweight, strong, and complex components. For instance, Boeing and Airbus have incorporated AM into their production lines to manufacture intricate parts for aircraft that reduce weight and improve fuel efficiency (Pant et al. 2021). Similarly, automotive giants like General Motors and Ford have adopted AM for prototyping and producing low-volume, custom parts, allowing for a quicker turnaround time in vehicle production (Yadav et al. 2021). In healthcare, AM has brought transformative changes to personalized medicine. 3D-printed prosthetics, implants, and even bio-printed tissues are increasingly being used to meet individual patient needs, offering tailored solutions that traditional manufacturing cannot achieve (Aimar et al. 2019). The technology also holds the potential for creating bio-printed organs and tissues, which could revolutionize organ transplantation in the future (Das et al. 2023). The healthcare sector's adoption of AM is expected to increase as research continues to improve the precision and biocompatibility of 3D-printed materials (Amaya-Rivas, et al. 2024). Despite its rapid growth, some challenges need to be addressed for AM to reach its full potential. These include issues related to material limitations, speed of production, surface finish quality, and the need for standardized processes. Furthermore, scaling up AM for large-volume manufacturing remains a significant hurdle (Mesecke et al. 2025). However, continued research and development are expected to overcome these obstacles, making AM more accessible for industrial-scale applications (Isania and Casalino 2023).

A comparative overview of additive manufacturing technologies

Additive manufacturing (AM) is a technology that builds three-dimensional objects by adding layers of material on top of each other. It's also known as 3D printing. This section provides a comparative overview of prominent AM technologies, highlighting their underlying principles, material suitability, and respective advantages and disadvantages. The discussion will encompass material extrusion, vat photopolymerization, powder bed fusion, and other emerging AM techniques (Fig. 4),(Table 1).

Material extrusion (Fused Deposition Modeling—FDM)

Fused Deposition Modeling (FDM), a widely adopted AM technique, involves the layer-by-layer deposition of heated thermoplastic filament. Its affordability and versatility make it a popular choice for prototyping and low-volume production. The process involves melting a thermoplastic filament and extruding it through a nozzle to create the desired three-dimensional structure. The simplicity of the process and the relatively low cost of the equipment contribute to its widespread adoption. However, FDM's limitations are also well documented. The mechanical strength and surface finish of FDM-produced parts are often inferior to those created by other AM technologies (Altıparmak et al. 2022; Fff 2022). This limitation stems from the inherent layer-by-layer nature of the process, which can result in weak interlayer bonds and a rough surface texture. Furthermore, the achievable dimensional accuracy is often limited by the resolution of the extrusion process and the thermal properties of the material. Recent research efforts are focused on

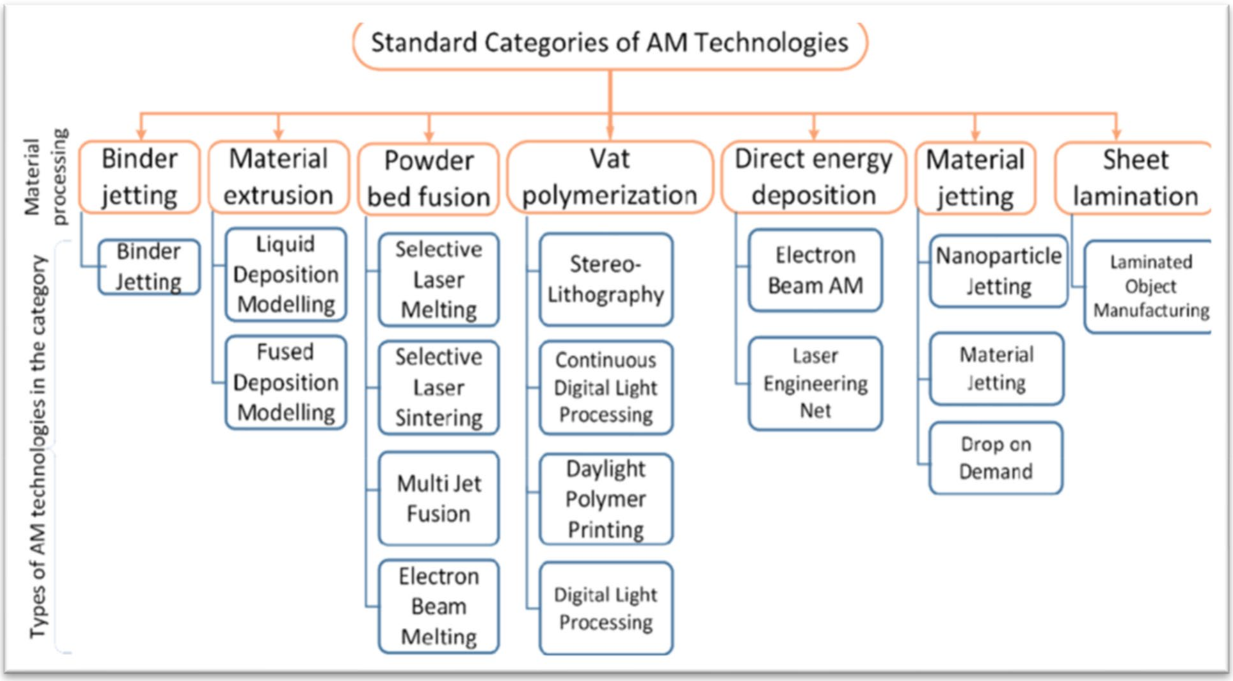

Fig. 4 The Seven (7) Standard Categories of AM Technology and their subcategory (Wakjira et al. 2021)

mitigating these limitations. Researchers are exploring the use of composite materials and recycled materials to enhance the mechanical properties of FDM-printed parts (Fff 2022; Pant et al. N.D.). Significant attention is also being paid to optimizing process parameters, such as layer thickness, extrusion speed, and nozzle temperature, to improve the overall quality and dimensional accuracy of the final products (Suniya et al. 2023). The ongoing research in this area is aimed at expanding the application range of FDM by addressing its inherent limitations and enhancing its capabilities (Fig. 5).

Vat photopolymerization (Stereolithography—SLA, Digital Light Processing—DLP)

Vat photopolymerization encompasses several AM techniques, most notably Stereolithography (SLA) and Digital

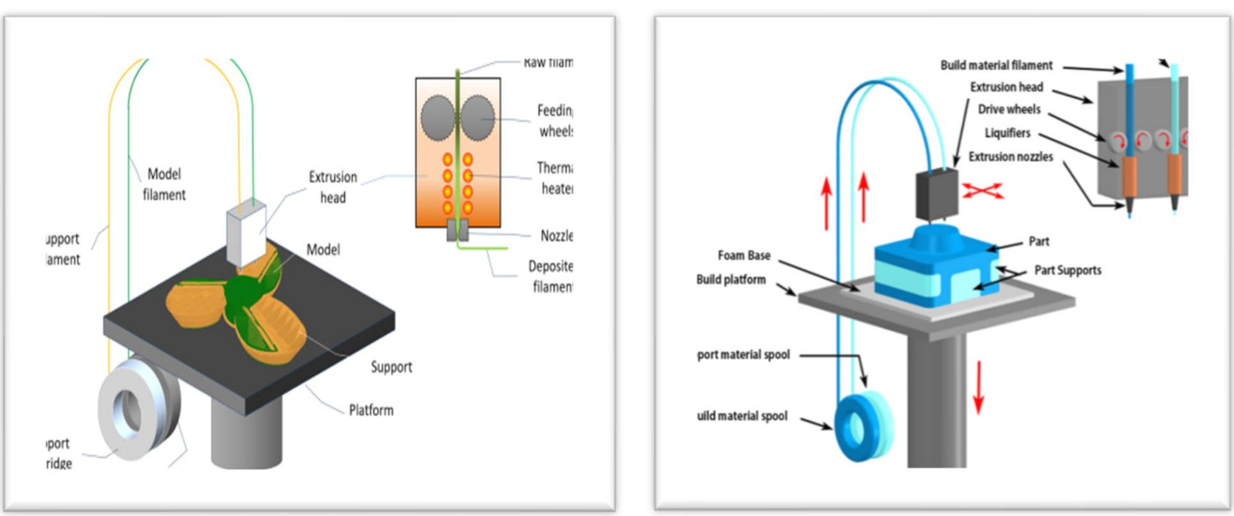

Fig. 5 Fused Deposition Modeling (FDM) (Dubey and Singh 2024)

Light Processing (DLP), that rely on the use of ultraviolet (UV) light to cure liquid photopolymer resin. These methods are capable of producing parts with exceptional precision and intricate geometries (Huang et al. 2023; Trends, et al. 2021; Al et al. 2022). The process involves projecting UV light onto a vat of liquid resin, causing the exposed areas to solidify. This layer-by-layer approach allows for the creation of complex shapes, including internal structures and overhanging features. SLA utilizes a laser to selectively cure the resin, while DLP uses a digital projector to cure a whole layer at once. This difference in light projection methods affects the speed and resolution of the process. The applications of SLA and DLP are diverse, ranging from dental prosthetics ("3D Printing/Additive Manufacturing Technologies in Dentistry : A Bibliometric Study Diş Hekimliğinde 3D Baskı/Katmanlı Üretim Teknolojileri 2025) and medical models to rapid prototyping and jewelry making. However, the material selection in vat photopolymerization is somewhat limited by the availability of suitable photopolymers. Furthermore, post-processing steps, such as cleaning, curing, and potentially surface finishing, are often necessary to achieve the desired properties and aesthetics of the final product (Bove et al. 2022). The development of new photopolymers with improved properties, along with advancements in post-processing techniques, continues to expand the capabilities and applications of these technologies (Fig. 6).

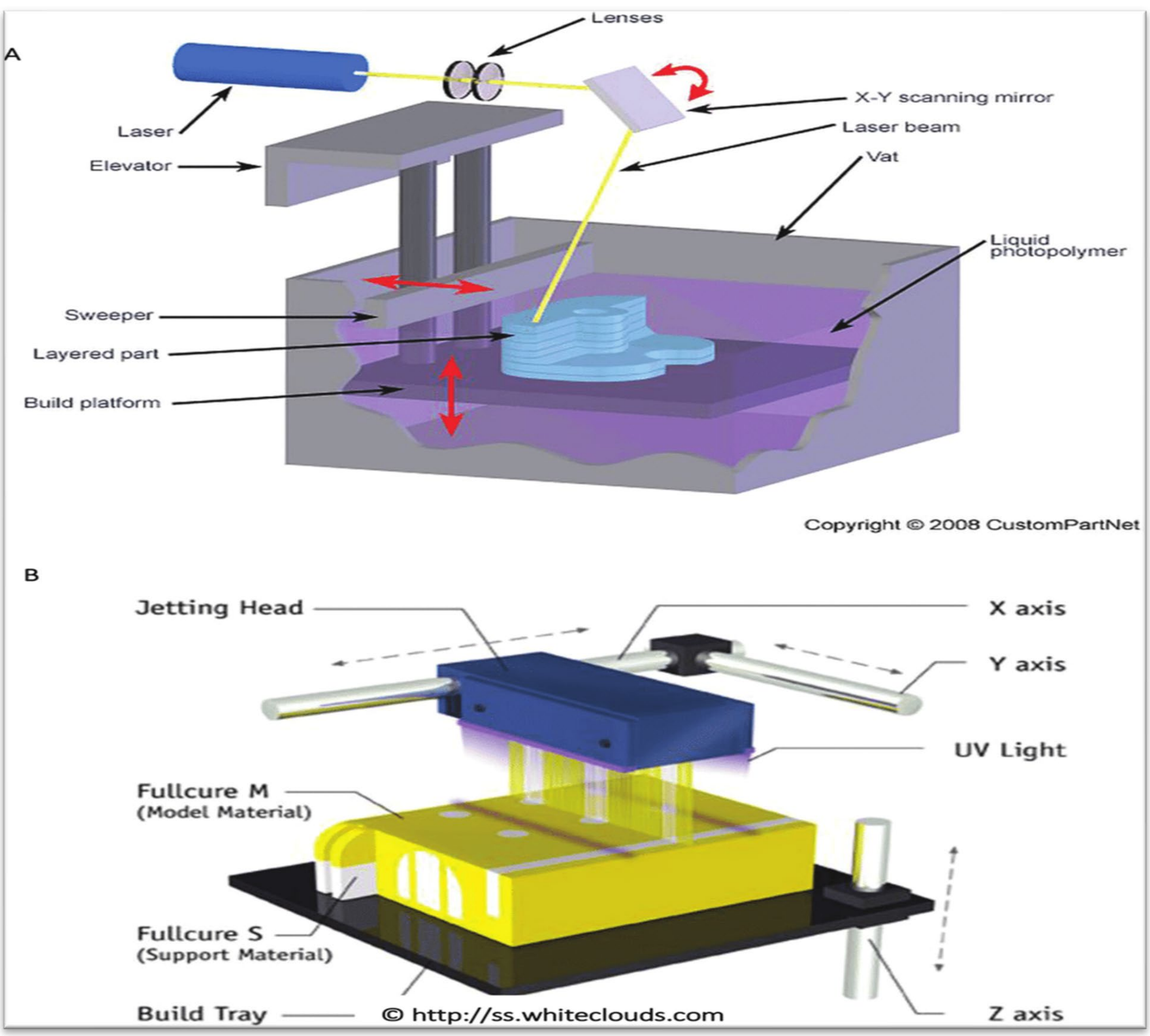

Fig. 6 **A)** Stereolithography (SLA), **B)** Digital Light Processing (DLP) (Ink 2022)

Powder bed fusion (Selective Laser Melting—SLM, Direct Metal Laser Sintering—DMLS, Electron Beam Melting—EBM)

Powder bed fusion (PBF) technologies represent a significant advancement in AM, particularly for metal components. These techniques involve the layer-by-layer melting and fusion of metallic powders using a focused energy source, typically a laser or an electron beam (Ink 2022; Istikorini and Sari 2022; Sharon et al. 2021). Selective Laser Melting (SLM), Direct Metal Laser Sintering (DMLS), and Electron Beam Melting (EBM) are prominent examples of this category. These processes offer high precision and the capability to produce highly complex metal parts with near-net shape accuracy (3D Printing/Additive Manufacturing Technologies in Dentistry : A Bibliometric Study Diş Hekimliğinde 3D Baskı/Katmanlı Üretim Teknolojileri 2025; Istikorini and Sari 2022; Mouzakis 2018). The ability to create intricate internal structures and complex geometries makes them particularly attractive for applications where traditional manufacturing methods are limited. However, PBF technologies are not without their challenges. The high initial investment costs associated with the equipment, the relatively slow build speeds, and the potential for residual stresses and porosity in the final parts limit their widespread adoption (Istikorini and Sari 2022; Sharon et al. 2021). Furthermore, the selection of appropriate process parameters and the careful control of the powder bed environment are crucial for ensuring the quality and consistency of the manufactured components (Jafari and Wits 2018). Ongoing research in this area focuses on enhancing process efficiency, improving material properties, and developing more robust quality control methods to make PBF technologies more accessible and reliable (Fig. 7).

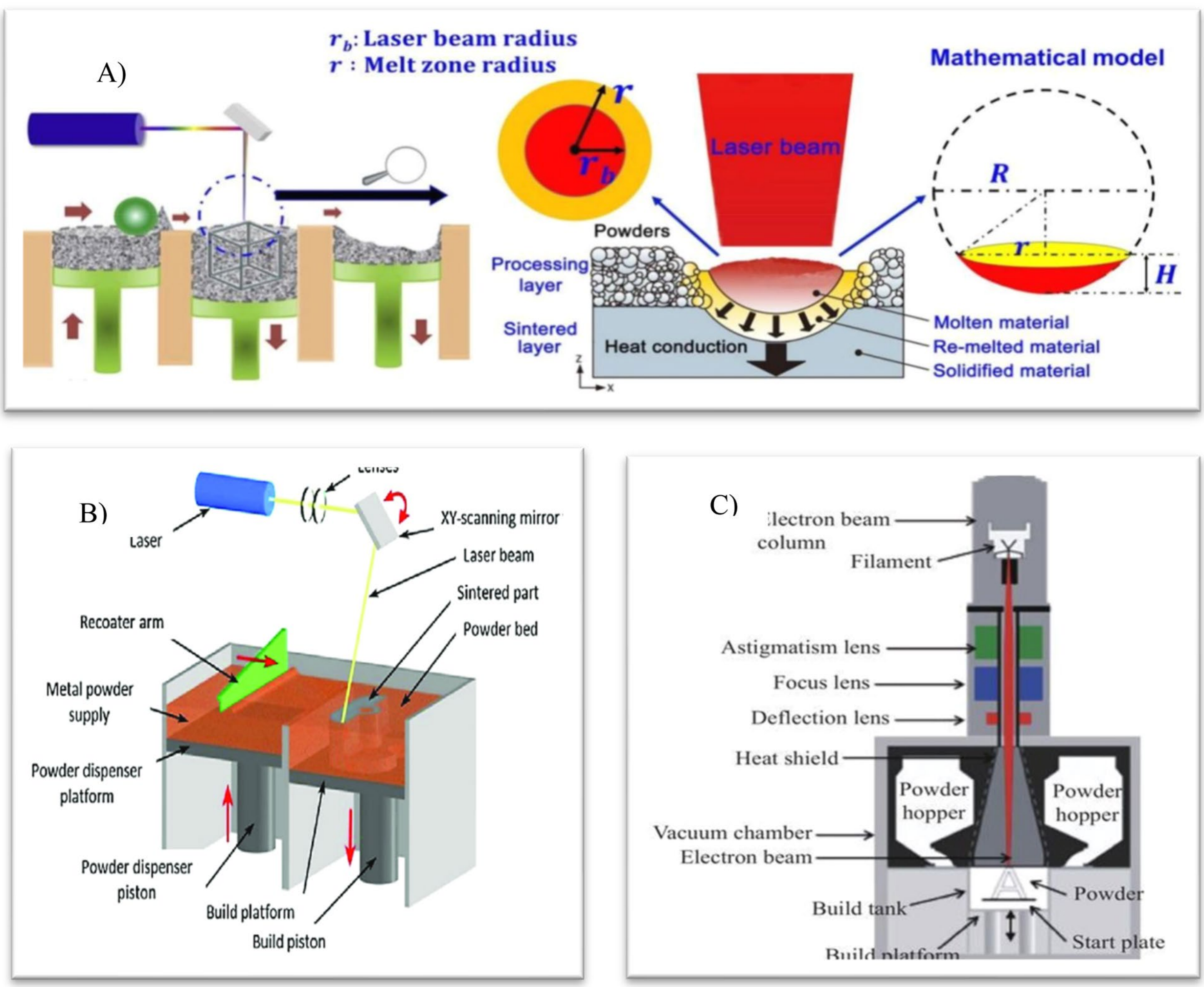

Fig. 7 **A)** Selective Laser Melting (SLM), **B)** Direct Metal Laser Sintering (DMLS), **C)** Electron Beam Melting (Mouzakis and Academy 2018)

Other AM technologies

Beyond the aforementioned technologies, several other AM techniques are gaining traction in various industries. These include material jetting, binder jetting, and directed energy deposition (DED) (Sharon et al. 2021; Steel et al. 2021). Material jetting involves the precise deposition of droplets of material, enabling high-resolution printing. Binder jetting, on the other hand, uses a binder to selectively join powder particles, suitable for large-scale production. Directed energy deposition (DED), including Wire Arc Additive Manufacturing (WAAM), stands out for its high deposition rates and the ability to produce large, complex components. DED processes involve melting and depositing material layer by layer using a focused energy source, such as a laser or an electron beam. WAAM, a specific DED technique, utilizes a wire feedstock as the material source, which is melted by an electric arc and deposited onto the build platform. The high deposition rates and the ability to create large components make WAAM particularly attractive for industrial applications. However, each of these methods has its own set of limitations. Material jetting, while offering high resolution, can be relatively slow and expensive. Binder jetting, although capable of high throughput, may require subsequent debinding and sintering steps. DED processes, while offering high deposition rates, can lead to challenges in controlling the melt pool dynamics and ensuring consistent material properties. The ongoing development and refinement of these technologies are expanding their capabilities and applications across various industries (Steel et al. 2021) (Table 1).

Table 1 Classification of additive manufacturing technologies and their general features (Wakjira et al. 2021; Altıparmak et al. 2022; Fff 2022; Pant et al. N.D.; Suniya et al. 2023; Dubey and Singh 2024; Huang et al. 2023; Trends, et al. 2021; Al et al. 2022; Bove et al. 2022; Ink 2022; Istikorini and Sari 2022; Sharon et al. 2021; Mouzakis 2018; Jafari and Wits 2018; Mouzakis and Academy 2018; Steel et al. 2021**)**

AM Technology	*Principle*	*Advantages*	*Limitations*	*Common Materials*	*Applications*
Material Extrusion (FDM/FFF)	A thermoplastic filament is heated and extruded through a nozzle to build parts layer by layer	-Low cost -Easy to use -Suitable for rapid prototyping	-Limited resolution - Anisotropic mechanical properties—Surface finish may require post-processing	PLA, ABS, PETG, TPU	Prototyping, educational models, and low-strength functional parts
Vat Photopolymerization (SLA/DLP)	A UV light cures liquid photopolymer resin layer by layer in a vat	-High resolution -Smooth surface finish -Suitable for intricate details	-Brittle parts - Limited material choices - Post-curing required	Photopolymer resins	Dental models, jewelry, microfluidics
Material Jetting (MJ)	Droplets of photopolymer are deposited and cured by UV light	-High accuracy -Multi-material and color-printing -Smooth surfaces	-Expensive - Limited mechanical properties - Material waste	Photopolymers, waxes	Prototypes, anatomical models, tooling
Binder Jetting (BJ)	A liquid binder selectively joins powder particles layer by layer	-Fast printing - No support structures needed - Wide material range	-Low part strength - Requires post -processing - Porosity issues	Sand, ceramics, metals	Casting molds, architectural models
Powder Bed Fusion (PBF)	A laser or electron beam fuses powder particles in a bed layer by layer	-High-strength parts—Complex geometries-Good mechanical properties	-High cost - Residual stresses - Limited build size	Titanium alloys, stainless steel, aluminum	Aerospace components, medical implants
Directed Energy Deposition (DED)	Focused thermal energy melts materials as they are deposited	-Repair and add features to existing parts-High deposition rates-Material flexibility	-Surface finish issues - Limited resolution - Complex equipment	Stainless steel, Inconel, titanium	Repair of aerospace parts, large structures
Sheet Lamination (LOM)	Sheets of material are bonded and cut layer by layer	-Low material cost - High speed - Large parts possible	-Limited to simple geometries - Poor surface finish - Limited material choices	Paper, metal foil, plastic sheets	Packaging prototypes, visual models
Hybrid Additive Manufacturing	Combines multiple AM processes or AM with subtractive methods	-Enhanced part properties - Multi-material capability - Improved functionality	-Complex process control—High equipment cost- Integration challenges	Various metals and polymers	Aerospace, automotive, and tooling

Materials in additive manufacturing,properties and selection

Material selection in AM is a critical factor influencing both the manufacturability of a part and its ultimate properties (Kawalkar et al. 2022; Razavi, et al. 2021; Elkanayati et al. 2023). A wide range of materials are employed in AM, including polymers, metals, ceramics, and composites. Polymers offer versatility, cost-effectiveness, and ease of processing, making them suitable for rapid prototyping and various applications. Metals, particularly in powder bed fusion processes, enable the creation of high-strength, durable components for demanding applications (Kawalkar et al. 2022). Ceramics, known for their high temperature resistance and hardness, find use in specialized applications. Composites, combining the properties of different materials, allow for tailored performance characteristics (Razavi, et al. 2021; Elkanayati et al. 2023). The development of AM-specific materials is an active area of research, focusing on overcoming limitations such as low strength, anisotropy, and poor surface finish. In biomedical applications, biocompatible materials are essential, requiring careful consideration of biodegradability, toxicity, and interaction with living tissues (Bikiaris, et al. 2023; Wu et al. 2022; Rothbauer et al. 2022; Tariq, et al. 2023). Furthermore, the growing emphasis on sustainability is driving the exploration of bio-based and recycled materials, aiming to reduce the environmental footprint of AM (Wakjira et al. 2021; Xu et al. 2018; Efstratiadis and Michailidis 2022). The selection of materials and their properties are central to the successful implementation of AM across various sectors, Fig. 8 (Table 2).

Applications of additive manufacturing across industries

Additive manufacturing, also known as 3D printing, has many applications across industries, including aerospace, energy, automotive, medical, and consumer goods. The versatility of AM has led to its adoption across a wide range of industries, each leveraging its unique capabilities to address specific manufacturing challenges. Application of additive manufacturing and its distribution over business sectors are shown in (Fig. 10).

Aerospace

AM has significantly impacted aerospace manufacturing, enabling the creation of lightweight, high-performance components with complex geometries (Blakey-milner et al. 2021; Giddaluri, et al. 2022; Gu et al. 2012). This technology allows for the production of parts that would be impossible or prohibitively expensive to manufacture using traditional methods. The reduction in weight

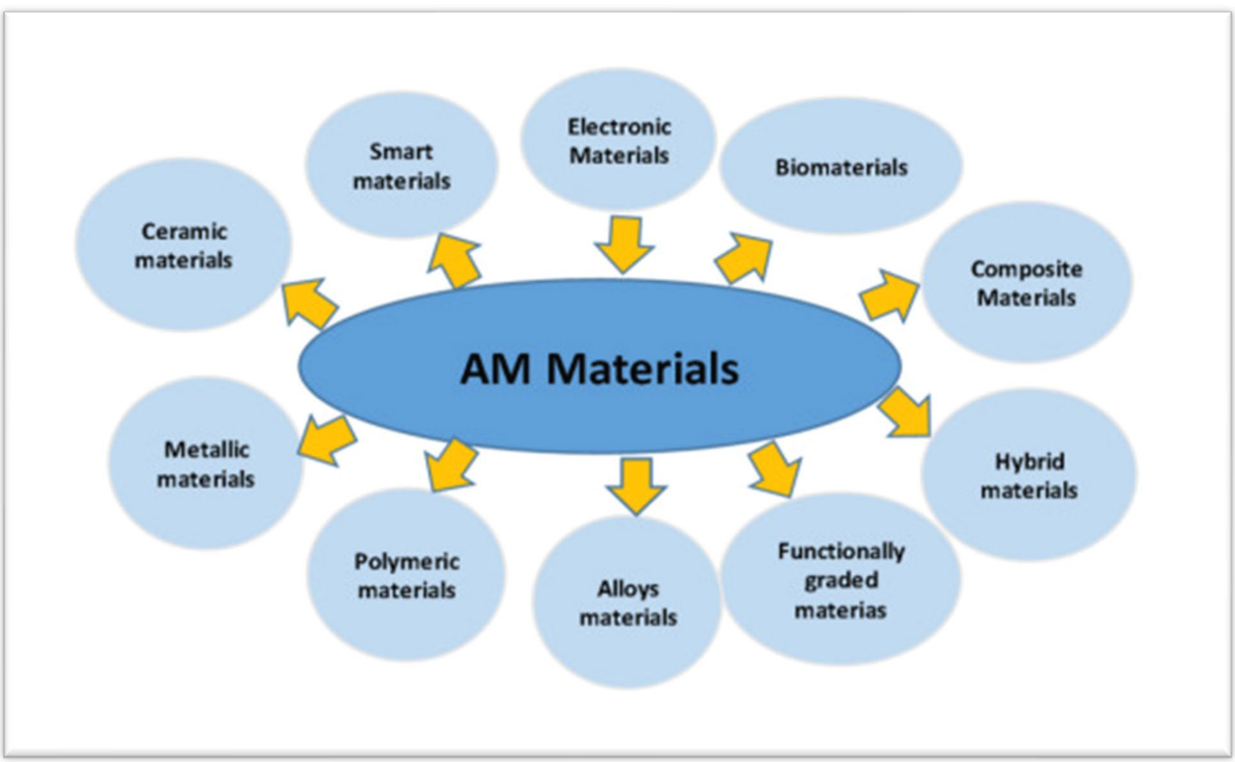

Fig. 8 Types of Materials in Additive Manufacturing (AM) (Srivastava et al. 2022)

Table 2 Here is a tabulated summary of materials commonly used in additive manufacturing (AM), along with their properties and selection considerations

Material Type	*Examples*	*Key Properties*	*Typical AM Process*	*Selection Criteria*	*Refe*
Polymers	PLA, ABS, PETG, Nylon	Lightweight, corrosion-resistant, moderate strength	Fused Deposition Modeling (FDM), SLS	Cost-effective, ease of printing, biodegradability (PLA)	Acierno and Patti 2023; Salifu et al. 2022)
Photopolymers	Epoxy, Acrylates	High surface finish, UV curable, brittle	Stereolithography (SLA), DLP	Precision applications, prototyping	Melentiev et al. 2024; Essmeister et al. 2022)
Metals	Ti6Al4V, SS316L, AlSi10Mg, Inconel 718	High strength, thermal stability, corrosion resistance	Selective Laser Melting (SLM), EBM, DED	Aerospace, biomedical, and energy	Yin et al. 2023; Blakey-Milner, et al. 2021)
Ceramics	Alumina, Zirconia, Silicon carbide	High hardness, heat resistance, brittle	Binder Jetting, SLA with ceramic slurry	Biomedical implants, electronics	Karamzadeh et al. 2025; Ly et al. 2021; Chen et al. 2022)
Composites	Carbon fiber-reinforced polymers (CFRP), Metal Matrix Composites (MMC)	High strength-to-weight ratio, custom properties	FDM, SLS, DED	Structural parts, aerospace, automotive	Adil and Lazoglu 2022; Jiang et al. 2021)
Bio-materials	Hydroxyapatite, Gelatin, Collagen, and PLA-based blends	Biocompatible, biodegradable, tissue compatibility	FDM, SLA, Bioprinting	Medical implants, tissue scaffolds	Kalyan and Kumar 2022; Sezer et al. 2021)
Smart Materials	Shape Memory Alloys (SMAs), Piezoelectrics, Magnetostrictive polymers	Responsive to stimuli (temperature, electricity)	SLM, Inkjet, DIW	Sensors, actuators, and self-healing devices	Lee Dec. 2022), Bhandari et al. 2024; Chen, et al. 2020)
Sustainable Materials	Recycled plastics, bio-based PLA, lignin blends	Eco-friendly, renewable, biodegradable	FDM, SLA	Sustainable manufacturing, low-cost applications	Kokare et al. 2023; Lodha et al. Nov. 2023)

translates to fuel efficiency and improved aircraft performance. Complex internal cooling channels and optimized structural designs can be incorporated directly into the components, enhancing their functionality and reducing material usage. Applications range from aircraft engines and fuselages to satellite components (Blakey-milner et al. 2021; Giddaluri, et al. 2022). The ability to produce customized parts on demand reduces lead times, simplifies supply chains, and enables efficient repairs (Khorasani et al. 2021). The development of in-space AM holds immense potential for on-site repairs and manufacturing, revolutionizing space exploration and reducing reliance on ground-based supply chains (Blakey-milner et al. 2021). This sector's adoption of AM underscores its potential for creating high-value, high-performance components in a demanding industry.

Biomedical

The use of AM in biomedical applications is experiencing exponential growth. AM enables the creation of patient-specific implants, prosthetics, and surgical guides, tailored to the individual's anatomy and needs (Hoffmann 2023). The ability to create intricate designs and porous structures promotes better integration with surrounding tissues and improves bone ingrowth. 3D bioprinting, a specialized subset of AM, allows for the fabrication of tissue constructs and organoids, offering unprecedented possibilities for regenerative medicine (Wu et al. 2022). However, challenges remain in ensuring the biocompatibility, sterilization, and long-term performance of AM-produced biomedical devices (Bikiaris, et al. 2023). The development of biocompatible materials, the optimization of printing parameters, and the rigorous testing of bioprinted structures are critical for ensuring the safety and efficacy of these applications. This field is at the forefront of AM's potential to revolutionize healthcare (Fig. 9).

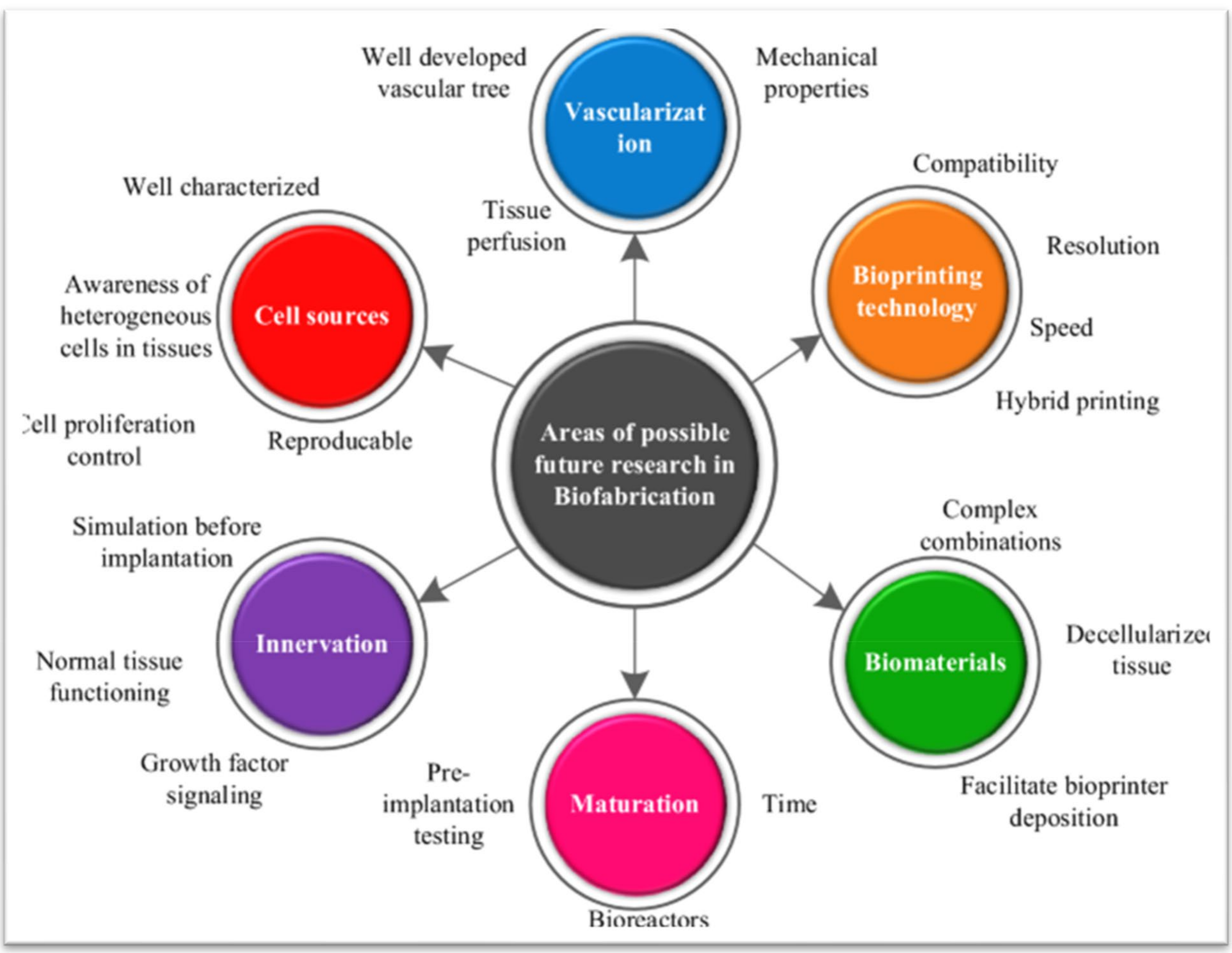

Fig. 9 Challenges faced by AM for biomedical applications (Ngo et al. Jun. 2018)

Construction

AM is emerging as a transformative technology in the construction industry, offering the potential to significantly improve efficiency and sustainability (Elkanayati et al. 2023; Directions 2022; Camacho et al. 2018). The ability to create complex geometries and customized components reduces material waste, labor costs, and construction time (Directions 2022). AM can enable the fabrication of large-scale structural elements, as well as intricate architectural features, previously unattainable with traditional methods. However, the scalability of AM in construction remains a significant challenge. The development of larger-scale AM systems, the availability of suitable construction materials, and the establishment of appropriate regulatory frameworks are essential for widespread adoption (Blakey-milner et al. 2021). The integration of AM with digital design tools and robotic systems further enhances the potential for automation and improved productivity. This application of AM promises to reshape the future of construction.

Other industries

The applications of AM extend far beyond aerospace, biomedical, and construction. It has found a place in numerous sectors, including automotive, consumer goods, and tooling (Dehghan et al. 2025; Haleem and Javaid 2019b; Felbrich et al. 2022). AM enables mass customization, allowing for the production of personalized products tailored to individual preferences. Rapid prototyping capabilities accelerate product development cycles, reducing time-to-market. On-demand manufacturing reduces inventory costs and minimizes waste. In tooling, AM facilitates the creation of complex molds and fixtures, improving efficiency and reducing production costs (Haleem and Javaid 2019b). The potential for sustainable manufacturing through material reuse and reduced waste is increasingly being explored (Ink 2022). The versatility and adaptability of AM are driving its expansion into new and diverse applications (Fig. 10).

Challenges and limitations of additive manufacturing

Despite the substantial progress and widespread adoption of AM, several challenges and limitations hinder its full potential (Razavi, et al. 2021; Dyl 2020). High initial investment costs for AM equipment and relatively slow production rates currently limit its adoption in high-volume manufacturing applications (Dyl 2020). Material limitations, the relatively small selection of materials compared to traditional manufacturing, and the inherent anisotropic properties of AM-produced parts affect their mechanical properties and reliability (Dehghan et al. 2025; Kawalkar et al. 2022). Surface roughness, residual stresses, and potential porosity in the final product can impact its performance and durability. The lack of standardized testing procedures and quality control methods poses a barrier to broader industrial acceptance (Jung et al. 2023; Mani et al. 2014). Furthermore,

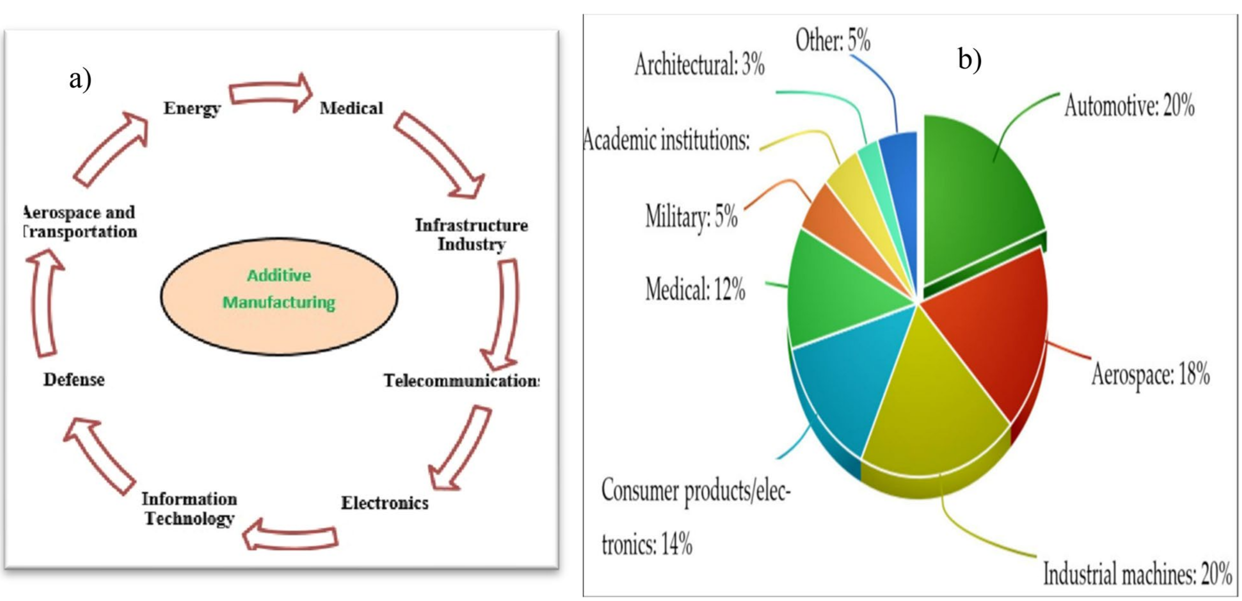

Fig. 10 **a**) Applications of additive manufacturing (AM), **b**) Distribution of additive manufacturing revenues over business sectors (Ink 2022; Vairagade et al. 2022)

Table 3 Advantages and Disadvantages of Additive Manufacturing (Baumann et al. Jan. 2018; Duruaku et al. 2023)

Technology	*Advantages*	*Disadvantages*
Material Extrusion	- Widespread, inexpensive- ABS plastic usable (good structural properties and accessible)- Suitable for visual models and prototypes	- Nozzle radius limited- Low accuracy and speed- Requires constant pressure of material
Powder Bed Fusion	- Integrates into small-scale systems- Office-sized machines- Powder support structure- Wide material options	- Relatively slow- Lacks structural material properties- Size limitations- High power usage- Finish dependent on powder grain size
Binder Jetting	- Multiple colors- Wide material range- Fast processing- Allows two-material use	- Not ideal for structural parts due to binder- Long post-processing time
Material Jetting	- High accuracy- Low material waste- Multiple materials and colors in one process	- Support structures often required- Limited material availability
Directed Energy Deposition	- High-quality, functional parts- High accuracy	- May require post-processing- Limited materials
Sheet Lamination	- High speed- Low cost- Easy material handling- Good accuracy and finish	- Limited materials- Fusion processes need more research
Vat Photopolymerization	- High accuracy- Fast process- Suitable for large build volumes	- Relatively expensive- Long post-processing time- Limited materials- Needs support structures

the environmental impact of AM, including energy consumption and material waste, needs careful evaluation and mitigation strategies (Istikorini and Sari 2022; Directions 2022). Addressing these challenges is essential for the continued growth and broader applicability of AM (Table 3).

The role of machine learning in additive manufacturing

Machine learning (ML) is playing an increasingly crucial role in addressing many of the challenges associated with AM and unlocking its full potential. ML algorithms can be effectively employed for process monitoring,

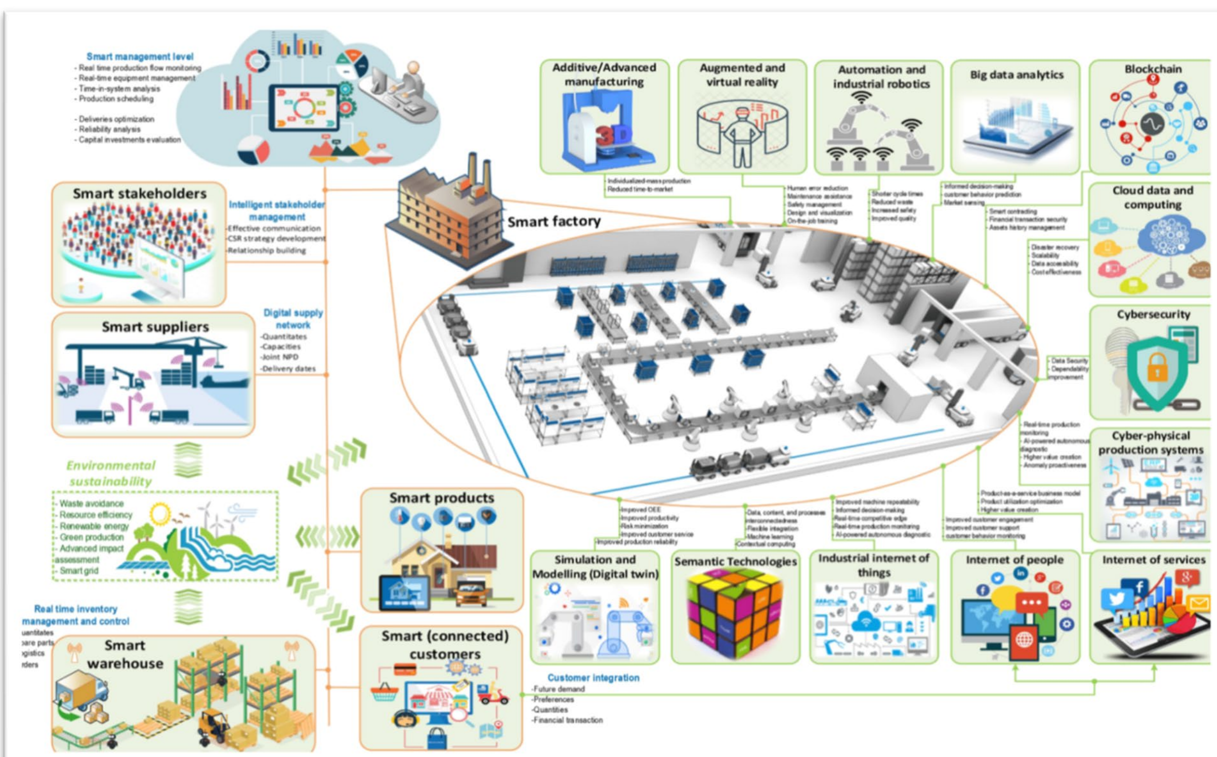

Fig. 11 The Role of Machine Learning in Additive Manufacturing (Wu 2023)

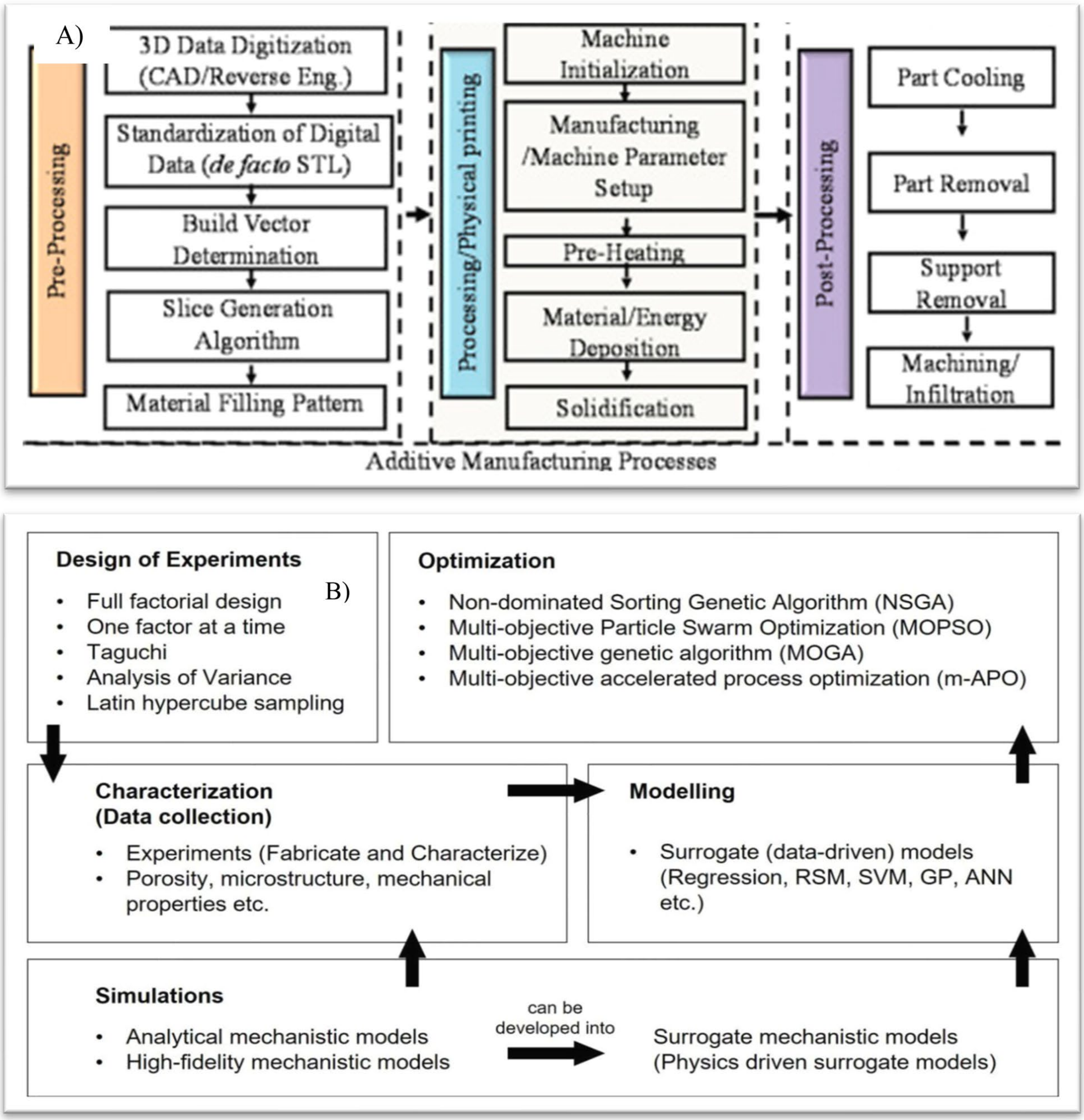

Fig. 12 **A)** Additive manufacturing process, **B)** Process optimization in additive manufacturing (Chia et al. 2022)

defect detection, design optimization, and material selection (Baumann et al. Jan. 2018; Qin et al. 2021). Process monitoring involves using sensors to collect data during the AM process, which is then analyzed by ML algorithms to identify anomalies and potential defects. Defect detection utilizes ML to identify and classify defects in manufactured parts, allowing for timely intervention and improved quality control. Design optimization employs ML to improve the design of parts for AM, taking into account the process capabilities and material properties. Material selection leverages ML to predict the best material for a given application and AM process. Sensor

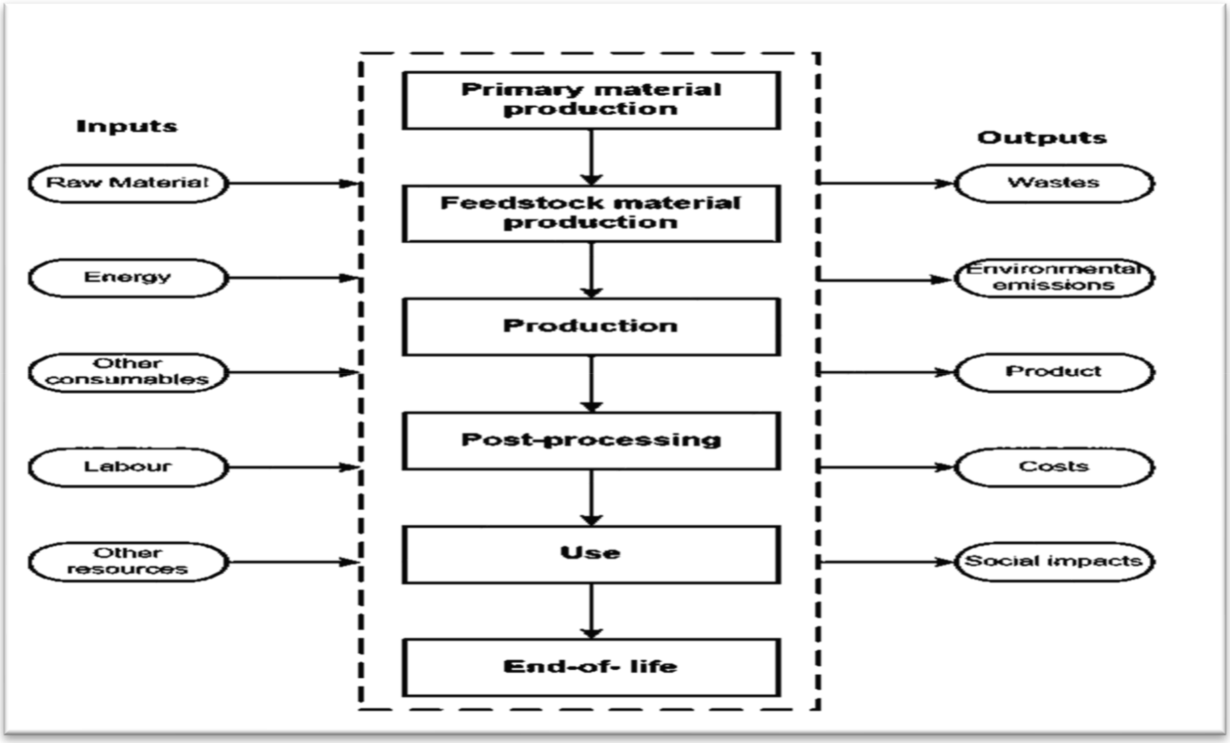

Fig. 13 Life Cycle Assessment Process of Additive Manufacturing (Kokare et al. 2023)

Table 4 Tabulated summary of recent advancements in additive manufacturing

Area of Focus	*Key Contribution*	*Summary*	*Refe*
Material Innovation	Nanocomposite filaments for FDM	Enhanced mechanical strength and thermal stability using CNT-infused filaments	Roque and Carlos 2021; Puppi and Chiellini 2020)
	Functionally graded materials in DED	Fabricated customized implants with gradient mechanical properties	Li et al. 2022)
Process Monitoring	In-situ defect detection in SLM	Integrated sensor systems for real-time defect mapping and melt pool monitoring	Li and Qu Apr. 2014)
	AI-based layer correction	Used deep learning to reduce distortion and improve dimensional accuracy	Hassan et al. 2024)
Sustainability	Life Cycle Assessment (LCA) of AM vs traditional methods	Found AM reduces waste, but some methods remain energy-intensive	Nguyen et al. Dec. 2024)
	Biodegradable polymers for bioprinting	Developed sustainable polymers for use in medical scaffolding	Soori et al. 2023)
Biomedical Applications	Vascularized bone scaffolds	Combined inkjet and FDM printing for complex tissue-compatible structures	Hisham and M. S. C, C. Dileep, L. Jacob, and H. Butt 2024)
Aerospace Applications	Lightweight aerospace components via EBM	Produced titanium parts, reducing aircraft weight and fuel consumption	Whenish et al. 2021)
AI & Industry 4.0 Integration	AI prediction of defects and optimization	Leveraged neural networks for pre-print defect detection and process simulation	Lin, et al. 2021)
	AM in Industry 4.0 systems	Discussed real-time AM integration in smart factories	Brion and Pattinson 2022)

Table 5 Below are the methodologies, parameters used, and major findings in recent additive manufacturing studies

Focus Area	Technique /Methodology	Key Parameters or Tools Used	Major Findings	Refe
Nanocomposite FDM Materials	Fabrication of polymer-CNT nanocomposite filaments	FDM at 200–230°C; CNT concentration: 0.5–2 wt.%	Enhanced tensile strength by 32%, improved thermal stability	Kumar et al. 2023)
Biomedical FGMs	Direct Energy Deposition (DED) of graded Ti6Al4V-hydroxyapatite	Laser power: 600–1000W; Gradient %: 10–50	Improved interface bonding, bioactivity for implants	Yu et al. 2014)
In-situ SLM Monitoring	Machine vision + thermal sensors for defect detection	Layer thickness: 30–50 µm; Sensor refresh rate: 100 Hz	Real-time porosity prediction accuracy ~ 92%	Ciccone et al. 2023)
AI-based AM Control	Deep CNNs for adaptive print correction	Input: layer-wise images; Output: G-code modifications	Reduced residual stress and improved layer alignment	2023)
Life Cycle Analysis	Comparative LCA for AM vs. subtractive CNC	Metrics: CO_2 emissions, material waste, energy (MJ/kg)	FDM showed 40% material savings, but 18% more energy use	Ali N.D.)
Sustainable Biopolymers	Development of PLA-alginate biocomposites for bioprinting	Printing temp: 190–210°C; Alginate ratio: 5–20%	Biodegradable with 25% higher cell adhesion	Santos and Santos 2024)
Biomedical Scaffold Printing	Hybrid inkjet + FDM bioprinting	Scaffold pore size: 300 µm; Layer height: 150 µm	Achieved vascularized bone scaffolds with cell viability > 85%	Sathish, et al. 2022)
Aerospace Metal Printing	Electron Beam Melting (EBM) of Ti6Al4V	Beam current: 20–30 mA; Build rate: 50–80 mm^3/s	Produced lightweight aerospace parts with < 2% porosity	Prashar et al. 2023)
AI in AM	AI/ML for defect prediction and path optimization	ANN & XGBoost trained on G-code and thermal data	Accuracy > 90% for defect classification	Jayawardane et al. 2023)
AM in Industry 4.0	Digital twin integration in AM workflows	Simulations synced with real-time AM sensor data	Improved predictive maintenance and remote process control	Rajora 2022)

Three-dimensional printing in construction (Sepasgozar et al. 2020)

fusion techniques, combined with ML, enable real-time process control and improved part quality (Akhavan and Manoochehri 2022; Kruth et al. Jan. 2007; Errico et al. 2024; Ho et al. 2021). The integration of ML into AM workflows promises to significantly improve efficiency, accuracy, and reliability, leading to wider adoption of this transformative technology. The synergy between ML and AM represents a powerful combination for advancing manufacturing capabilities (Fig. 11).

Process optimization in additive manufacturing

Process optimization plays a crucial role in improving the efficiency and quality of AM processes. By adjusting key parameters such as print speed, layer thickness, and energy input, manufacturers can enhance material properties, reduce defects, and minimize costs. AI-driven models and real-time monitoring systems further aid in optimizing print conditions to achieve consistent and high-quality outputs (Fig. 12).

Life cycle assessment of additive manufacturing

Life Cycle Assessment (LCA) evaluates the environmental impact of AM processes, from raw material extraction to end-of-life disposal. Factors such as energy consumption, material waste, emissions, and recyclability are analyzed to determine the sustainability of AM compared to conventional manufacturing methods. Improving LCA outcomes involves developing eco-friendly materials, enhancing energy efficiency, and implementing circular economy principles (Fig. 13).

Discussion on recent advancements in additive manufacturing

Additive manufacturing (AM) has undergone significant transformations in the last five years, influenced by theoretical frameworks in thermal behavior, material performance, and design freedom. This section synthesizes and compares recent findings with established theories to provide a cohesive understanding of advancements in the field and these are described in Tables 4, 5 and 6, (Figs. 14 and 15).

Table 6 Here is a comparison of additive manufacturing and subtractive manufacturing

Aspect	*Additive Manufacturing*	*Subtractive Manufacturing*	*Refe*
Definition	Builds objects layer by layer from digital models, using materials like polymers, metals, or ceramics	Involves removing material from a solid block using machining, milling, drilling, etc	Sepasgozar et al. 2020; Ingarao and Priarone 2020)
Material Utilization	Efficient material usage with minimal waste	Higher material waste due to cutting/removal	Sepasgozar et al. 2020; Vafadar et al. 2021)
Complex Geometry	Capable of producing complex, intricate geometries	Limited to tool access and machining constraints	Ingarao and Priarone 2020; Simha et al. 2024)
Tooling Requirement	Minimal tooling; mainly requires the printer and material	Requires various cutting tools and fixtures	Sepasgozar et al. 2020; Lusicini et al. 2025)
Production Speed	Slower for large parts but better for prototyping	Faster for mass production and with simple parts	Vafadar et al. 2021; Chea et al. 2024)
Surface Finish	Often requires post-processing for a good finish	High precision and a smooth surface are achievable	Ingarao and Priarone 2020; Lusicini et al. 2025)
Material Types	Growing range, especially in polymers and metals	Wide range, including metals, plastics, and composites	Vafadar et al. 2021; Fang et al. 2024)
Environmental Impact	More sustainable with less waste and energy use	High energy use and material waste	Simha et al. 2024; Fang et al. 2024)
Cost Efficiency	Economical for low-volume, customized parts	Cost-effective for mass production	Sepasgozar et al. 2020; Lusicini et al. 2025)
Customization	Highly flexible or suited for customization and rapid prototyping	Less flexible or suited for design changes	Sepasgozar et al. 2020) (Fang et al. 2024)
Sustainability	More sustainable due to lower material waste	Generates more waste, energy-intensive	Wang and Fuh 2023; Huang et al. Feb. 2015)
Application Fields	Aerospace, biomedical, architecture, consumer goods	Aerospace, automotive, and tooling	Frölich, et al. 2024; Banerjee et al. 2025)
Automation and Integration	Integrating with AI and Industry 4.0 systems	High integration with CNC and robotics	Farhan Khan, et al. 2021; Al Rashid and Koç 2023)

Thermal behavior and process control

Thermal management is a foundational concept in AM, particularly for metal-based techniques like Selective Laser Melting (SLM) and Electron Beam Melting (EBM). According to thermal modeling theories (Ghanavati and Naffakh-moosavy 2021), rapid heating and cooling cycles induce thermal gradients that cause residual stress and distortion. Zhang et al. validated this through real-time thermal imaging and in-situ sensors in SLM processes, highlighting the correlation between thermal behavior and porosity formation (Patel and Taufik 2021). Their approach aligns with Kruth et al.'s earlier findings (Preisler and Str 2023) and builds upon them using convolutional neural networks (CNNs) for thermal anomaly detection. Similarly, Gao et al. ("PLA bio -nanocomposites reinforced with cellulose nanofibrils (CNFs) for 3D printing applications" 2023) implemented AI-based feedback systems to dynamically adjust laser power, reducing heat-induced warping, a practical realization of cyber-physical system theory in AM.

Material performance and functionally graded design

Traditional material theories, including functionally graded materials (FGMs) and heterogeneous design, have become central to enhancing AM applications (Gibson and Rosen N.D.). Recent developments focus on customizing material behavior spatially within a part. Li et al. (Heyraud, et al. 2023) extended these concepts by fabricating polymer nanocomposite filaments with gradient mechanical properties for Fused Deposition Modeling (FDM). Their study supports the FGMs framework by achieving variable stiffness and improved thermal stability in a single build. In a similar vein, Kumar and Bandyopadhyay (Melchels et al. 2010) explored direct energy deposition (DED) for metal FGMs used in biomedical implants, demonstrating seamless material transitions as a solution to interfacial stress issues previously identified in layered materials (Alex et al. N.D.). In the context of sustainability, Nguyen et al. (Yoder et al. 2018) utilized biodegradable PLA-cellulose blends, supporting Cradle-to-Cradle (C2C) material theory by enabling material

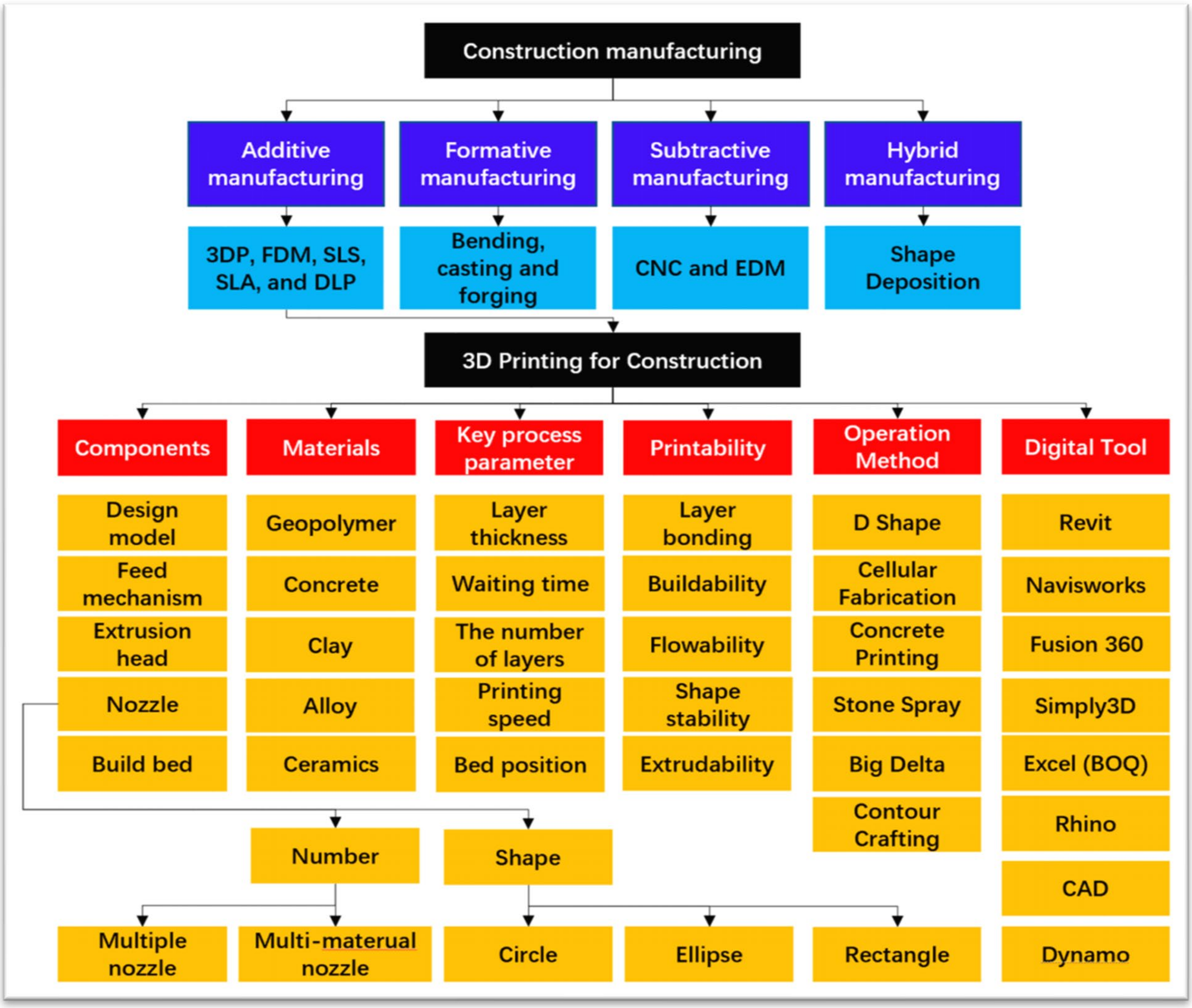

Fig. 14 Taxonomy of 3DP systems, digital tools, operation methods, and required materials for construction manufacturing

recovery and bio-integration, especially in biomedical AM.

Design freedom and customization

Design freedom central to the Design for Additive Manufacturing (DfAM) framework enables the creation of complex geometries, lattice structures, and customized forms (Kreuzer et al. 2024). AM's value proposition lies in shifting manufacturing from constraint-limited design to performance-driven design. In biomedical applications, Chen et al. (Tao et al. 2019) fabricated vascularized bone scaffolds using hybrid AM techniques, allowing both macro and micro-level customization, an embodiment of DfAM principles and topological freedom. This directly supports Melchels et al.'s (Fenta and Mebratie 2024) early claims on the anatomical precision of AM in regenerative medicine. For aerospace, Smith et al. (Rouf, et al. 2022) applied topology optimization to AM-fabricated titanium components, achieving mass reductions while maintaining structural integrity. This supports the lightweight design theory and fulfills structural performance requirements with fewer design constraints.

Artificial intelligence and industry 4.0 integration

Finally, AM is increasingly integrated with digital manufacturing via AI and digital twins, grounded in the Smart Manufacturing Framework. Singh and Rahman (Lin et al. 2023) reviewed predictive modeling techniques that simulate layer-wise stress distribution and

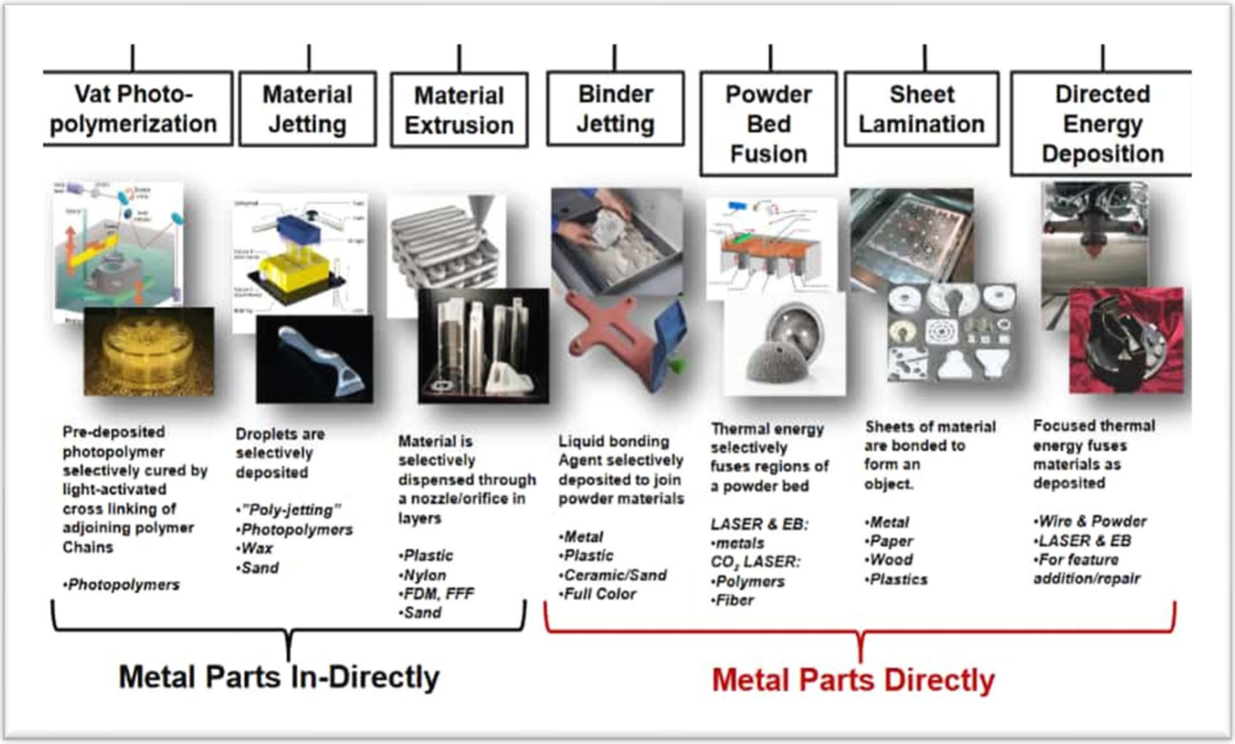

Fig. 15 Factors to Consider When 3D Printing or Additive Manufacturing Metal Parts (Kardys 2017)

adjust toolpaths dynamically. Their work echoes Tao et al.'s (Segovia-guerrero et al. 2025; Landi et al. 2023) concept of digital twins for real-time virtual replication and performance prediction in AM.

Future directions

The future of AM is characterized by ongoing research aimed at overcoming existing limitations and expanding its applications (Isania and Casalino 2023; Kawalkar et al. 2022; Razavi, et al. 2021). Advancements in materials science are crucial for developing new materials with enhanced properties, tailored for specific AM processes and applications (Wu 2023; Pant et al. N.D.). Process optimization will continue to focus on increasing build speeds, improving surface finish, reducing residual stresses, and enhancing the overall quality and reliability of AM-produced parts. The integration of ML and AI will play a pivotal role in automating and optimizing AM processes, leading to greater efficiency and improved product quality. Standardization efforts and lifecycle assessments are vital for ensuring the sustainable and responsible adoption of AM across industries (Amaya-Rivas, et al. 2024; Elkanayati et al. 2023; Bikiaris, et al. 2023). The convergence of AM with Industry technologies will further transform manufacturing paradigms, driving innovation and creating new opportunities (Dubey et al. 2024). This review has highlighted the significant progress made in AM while acknowledging the challenges that persist. Continued interdisciplinary research and collaboration among materials scientists, engineers, and computer scientists are essential to fully realizing the transformative potential of this technology. The future of AM is marked by a convergence of technological advancements and a growing focus on sustainability and responsible manufacturing (Kokare et al. 2024), Table 7, (Fig. 16).

Conclusions

Additive manufacturing (AM), also widely recognized as 3D printing, continues to redefine modern manufacturing through its capacity to fabricate complex geometries and enable rapid prototyping, significantly shortening product development cycles. With a diverse array of AM technologies ranging from fused deposition modeling (FDM) to selective laser sintering (SLS) and direct metal laser sintering (DMLS), material compatibility

Table 7 Future Directions and their description of Additive Manufacturing (AM)

Future Direction	***Description***	***Refe***
Integration of Artificial Intelligence (AI)	AI is increasingly being integrated into AM processes to enhance design optimization, predictive maintenance, and real-time quality control. This integration enables more efficient, reliable, and customized production. AI algorithms can analyze complex geometries and material properties to recommend optimized designs, predict equipment failures before they occur, and monitor production quality in real-time	Advancements 2024)
Digital Twins and Smart Factories & Optimization	The development of digital twin frameworks allows for real-time predictive control of AM processes. By simulating and optimizing process parameters, manufacturers can achieve desired mechanical properties and improve overall process efficiency. This approach leverages machine learning and Bayesian optimization techniques to predict and control temperature profiles during fabrication	Karkaria et al. 2024)
In-Situ Process Monitoring and Quality Control	Advanced in-situ monitoring techniques, including optical and acoustic sensors, are being employed to detect defects during the AM process. Coupled with machine learning algorithms, these systems enable real-time defect detection and adaptive quality enhancement, moving towards zero-defect manufacturing	Chen et al. 2024)
Cooperative Robotics in AM	The integration of cooperative robotics in AM facilitates the fabrication of large and complex structures. Multiple robotic arms can work in unison, enabling multi-material deposition and accelerating production speeds. This approach also enhances defect detection and correction through advanced sensing technologies	Rescsanski et al. 2025)
Sustainability and Circular Manufacturing	AM is contributing to sustainability through the use of recyclable and bio-based materials, energy-efficient processes, and closed-loop recycling systems. These practices reduce waste, lower energy consumption, and support environmentally friendly manufacturing	Kalyan et al. 2021)
On-Demand and Localized Production	The shift towards on-demand and localized production is transforming supply chains. By producing parts as needed and closer to the point of use, companies can reduce inventory costs, minimize transportation emissions, and respond more rapidly to market demands	Pose-rodriguez 2021)
Expansion Across Industries	AM is expanding beyond traditional sectors like aerospace and healthcare into industries such as construction, consumer goods, and food. This diversification is driven by AM's ability to produce complex, customized, and lightweight components efficiently	Kumar et al. 2021)
Advancements in Deep Learning Applications	Deep learning techniques are being applied to various aspects of AM, including design optimization, process modeling, and defect detection. These advancements enable more accurate predictions, improved process control, and the development of innovative designs	Islam et al. N.D.)
Multi-Material and Large-Format 3D Printing	Developments in multi-material and large-format 3D printing are enabling the production of parts with varying material properties and larger dimensions. This capability is particularly beneficial for applications requiring complex structures and diverse material characteristics	Chattopadhyay, et al. 2021)
Smart AM and Faster Printing Speeds	The integration of smart technologies, such as AI and IoT, is leading to faster and more efficient AM processes. These advancements result in reduced production times, lower costs, and enhanced product quality	Ng et al. 2024b)
Advancements in Material Science	Development of high-performance alloys, composites, and ceramics. Smart materials that respond to environmental stimuli (e.g., shape-memory alloys, self-healing polymers). Sustainable or eco-friendly materials (e.g., bio-based polymers, recycled composites). Improved material characterization techniques at the nano and micro scales, enabling better design and performance prediction	Chen, et al. 2025)
Mobile Additive Manufacturing	On-site part fabrication in remote or critical environments (e.g., space missions, military operations, disaster zones). Maintenance and repair operations in situ, eliminating the need to return components to centralized facilities. Use in construction (e.g., mobile concrete printers for buildings and bridges). Development of mobile AM units for on-site production, beneficial in remote or challenging environments	Gunasegaram, et al. 2023)
Hybrid Manufacturing Processes	Combining 3D printing (additive) with CNC machining (subtractive) to produce complex geometries with high surface finish. Integrating laser cladding (additive) with milling or grinding for tool repair or customization. Using direct energy deposition (DED) alongside traditional processes for rapid prototyping and component restoration	Gunasegaram, et al. 2023)

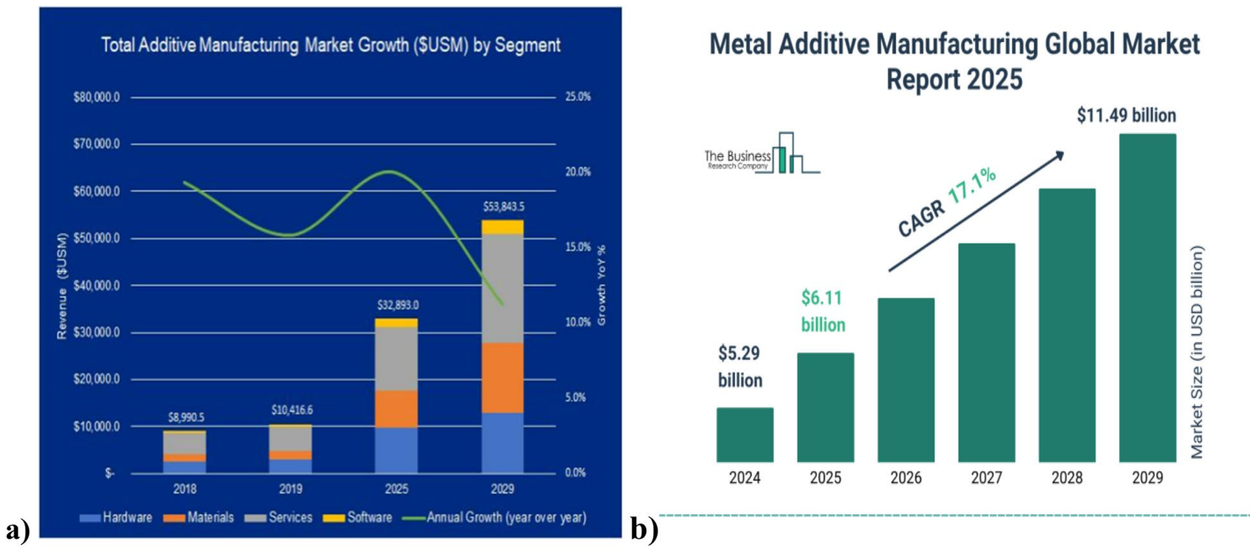

Fig. 16 **a)** Total additive manufacturing market growth segment, **b)** Metal additive manufacturing global market reports (SmartTech 2019)

and process efficiency have become central to selecting the optimal method for specific applications. Despite its advantages, several persistent challenges hinder widespread industrial adoption. These include limitations in surface finish quality, inconsistencies in mechanical properties, and high production costs for large-scale or high-performance components. Recent research emphasizes that AI-driven optimization techniques are increasingly being leveraged to address these issues, particularly in enhancing process control, automating defect detection, and optimizing material selection and print parameters.

Furthermore, sustainability has emerged as a critical focus in AM research, with significant efforts directed toward integrating recycled feedstocks, minimizing material waste, and adopting bio-based or biodegradable materials. Life cycle assessments (LCAs) of AM processes are being increasingly utilized to evaluate environmental impacts, offering valuable insights into the ecological viability of AM compared to traditional manufacturing. Overall, additive manufacturing is evolving into a highly intelligent and sustainable manufacturing platform, with AI integration and green material innovations playing transformative roles. Continued interdisciplinary research, particularly in machine learning, material science, and sustainability, will be pivotal to overcoming current barriers and unlocking the full industrial potential of AM.

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Data availability
All data that support the findings of this study are included within the article. If any supplementary files needed to support this study are available on request from the corresponding author, upon reasonable request.

Declarations

Competing interests
The authors declare no competing interests.

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